

PARTS LIST

Semiconductors:

BOARD1 - Digispark ATtiny85 microcontroller development board
 IRF1 - IRL520N MOSFET
 D1 - 1N5817 Schottky diode
 W.LED - White power LED for bike

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 100-ohm
 R2 - 10-kilo-ohm
 VR1 - 10-kilo-ohm pot

Capacitors:

C1 - 220 μ F electrolytic

Miscellaneous:

CON1, CON3 - 2-pin connector
 CON2 - 9-pin connector
 BATT.1 - 9V USB battery

of the 9V battery, the 2-pin connector CON1 can be used to connect its external charger to recharge the battery. It is worth noting that the nominal output of most rechargeable 9V LiPo batteries is only 7.4V (8.2V maximum). So, there is a possibility that the Digispark board will reset due to power drop when a high-power lamp is energised. The 220 μ F electrolytic capacitor C1 mitigates that episodic hiccup. Since Digispark has an onboard voltage regulator, you can safely use a 2S, 3S, or 4S LiPo battery pack instead of the proposed 9V USB battery.

There is no fixed rule on how many lumens of brightness you need for your bike light, but typically you will need approximately 500-lumen front light to see where you are going than to be just seen. So, use a pair of one-watt 'star white' LEDs in series as the light source, and try to tweak it later after some road trials. Include a suitable ballast resistor to limit the operating current, as shown in Fig. 4.

The LED lamp needs some optics to improve the light coming from it. Nowadays, LED reflector/lens plus a holder come together as a kit. However, while choosing it, check its 'beam angle' specs—the angle over which the light would be distributed. Narrow beam angles (for example 40°) have a tight beam of light and wide beam angles (around 120°) have a larger coverage for wide area lighting.

Also note that a clear lens provides a crisp-edged beam of light and a diffused lens offers a more uniform light output, especially when multiple lamps are used.

Warning. Sometimes strobe light-

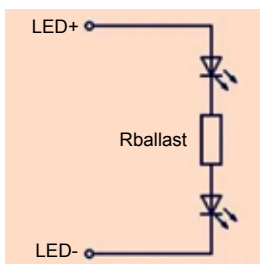


Fig. 4: Ballast resistor

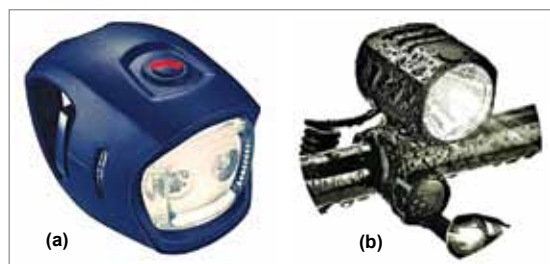


Fig. 5: Proposed enclosures (a) and (b)

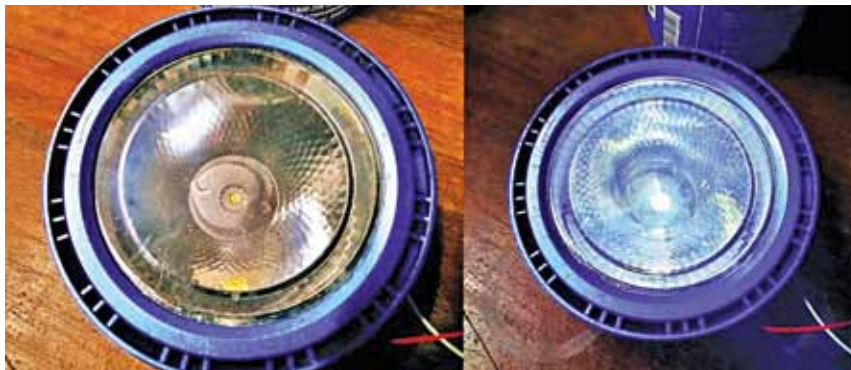


Fig. 6: White LED torch head for testing

ing can trigger seizures in persons suffering from photosensitive epilepsy (https://psychology.wikia.org/wiki/Photosensitive_epilepsy)!

Construction

If you are considering building a real-world model of this LED bike light, please use a rigid, waterproof enclosure. Some reference models are shown in Fig. 5.

For a quick functionality test, a large white LED torch head can be scavenged from an old torchlight to act as the light source, as shown in Fig. 6.

Software

The Arduino's pretty simple sketch (code) ported for Digispark is given here. It turns the lamp drive pin (P1) on and off with delays in between set by the pot connected to analogue input A1 (P2). In order to get much better strobes, however, a bit of complex math is applied in an interesting way to fine-tune the time delay function.

```
/* LED Bike Light v1
 * Key Hardware: Digispark Attiny85
 * Development Board
 * Arduino IDE: 1.8.13 (Windows10x64)
 * Author: T.K.Hareendran
 * Adapted Open-Source Code. Thanks to
 * www.bukovac.si */
int headLamp = 1;
int val = 0;
int onTime = 0;
int offTime = 0;
void setup() {
  pinMode(headLamp, OUTPUT);
}
void loop() {
  val = analogRead(1);
  onTime = val;
  offTime = (-2000000 / (onTime + 1000)) + 2000;
  onTime = onTime / 4;
  offTime = offTime / 4;
  if (onTime > 12) {
    digitalWrite(headLamp, HIGH); }
  delay(onTime);
  if (onTime < 250){
    digitalWrite(headLamp, LOW); }
  delay(offTime); } EFY
```

EFY Note

The source code of this project is available for free download at source.efymag.com



T.K. Hareendran is an electronics designer, hardware beta tester, author, and product reviewer